

Here is some intuition as to why one weight is negative in this example. It follows from Equation (7) in the proof of Theorem 1 (see also Theorem 1 in Goodman-Bacon, 2018) that in this simple example, $\beta_{fe} = (\text{DID}_1 + \text{DID}_2)/2$, with

$$\begin{aligned}\text{DID}_1 &= E(Y_{2,2}) - E(Y_{2,1}) - (E(Y_{1,2}) - E(Y_{1,1})), \\ \text{DID}_2 &= E(Y_{1,3}) - E(Y_{1,2}) - (E(Y_{2,3}) - E(Y_{2,2})).\end{aligned}$$

The first DID compares the evolution of the mean outcome from period 1 to 2 in group 2 and in group 1. The second one compares the evolution of the mean outcome from period 2 to 3 in group 1 and in group 2. The control group in the second DID, group 2, is treated both in the pre and in the post period. Therefore, under the common trends assumption, it follows from Lemma 1 in Appendix A (a similar result appears in Lemma 1 of de Chaisemartin (2011) and in Equation (13) of Goodman-Bacon (2018)) that $\text{DID}_1 = E[\Delta_{2,2}]$, but

$$\text{DID}_2 = E[\Delta_{1,3}] - (E[\Delta_{2,3}] - E[\Delta_{2,2}]).$$

DID_2 is equal to the ATE in group 1 period 3, minus the change in group 2's ATE between periods 2 and 3. Intuitively, the mean outcome of groups 1 and 2 may follow different trends from period 2 to 3 either because group 1 becomes treated, or because group 2's ATE changes. The intuition that negative weights arise because $\hat{\beta}_{fe}$ uses treated observations as controls also appears in Borusyak and Jaravel (2017).

We now generalize the previous illustration by characterizing the (g, t) cells whose ATEs are weighted negatively by β_{fe} .

Proposition 1 *Suppose that Assumption 1 holds and for all $t \geq 2$ $N_{g,t}/N_{g,t-1}$ does not vary across g . Then, for all (g, t, t') such that $D_{g,t} = D_{g,t'} = 1$, $D_{.,t} > D_{.,t'}$ implies $w_{g,t} < w_{g,t'}$. Similarly, for all (g, g', t) such that $D_{g,t} = D_{g',t} = 1$, $D_{g,.} > D_{g',.}$ implies $w_{g,t} < w_{g',t}$.*

Proposition 1 shows that β_{fe} is more likely to assign a negative weight to periods where a large fraction of groups are treated, and to groups treated for many periods. Then, negative weights are a concern when treatment effects differ between periods with many versus few treated groups, or between groups treated for many versus few periods.

Proposition 1 has interesting implications in staggered adoption designs, a special case of sharp designs defined as follows.

Assumption 6 (*Staggered adoption designs*) *For all g , $D_{g,t} \geq D_{g,t-1}$ for all $t \geq 2$.*

Assumption 6 is satisfied in applications where groups adopt a treatment at heterogeneous dates (see e.g. Athey and Stern, 2002). In that design, Borusyak and Jaravel (2017) show that β_{fe} is more likely to assign a negative weight to treatment effects at the last periods of the panel.

This result is a special case of Proposition 1: in staggered adoption designs, $D_{.,t}$ is increasing in t , so Proposition 1 implies that $w_{g,t}$ is decreasing in t .¹⁰ Proposition 1 also implies that in that design, groups that adopt the treatment earlier are more likely to receive some negative weights.

Finally, in staggered adoption designs, Athey and Imbens (2018) derive a decomposition of β_{fe} that resembles to, but differs from, that in Theorem 1. They derive their decomposition under the assumption that the dates at which each group starts receiving the treatment are randomly assigned, while we derive ours under a common trends assumption.

3.2 Robustness to heterogeneous treatment effects

Theorem 1 shows that in sharp designs with many groups and periods, $\hat{\beta}_{fe}$ may be a misleading measure of the treatment effect under the standard common trends assumption, if the treatment effect is heterogeneous across groups and time periods. In the corollary below, we propose two robustness measures that can be used to assess how serious that concern is.

Those robustness measures are defined conditional on $\mathbf{D}$, the vector stacking together the treatments of all the (g, t) cells. Specifically, for all $(g, t) \in \{1, \dots, G\} \times \{1, \dots, T\}$, let $\tilde{\Delta}_{g,t} = E(\Delta_{g,t} | \mathbf{D})$ denote the ATE in cell (g, t) conditional on $\mathbf{D}$,¹¹ let $\tilde{\Delta}^{TR} = E(\Delta^{TR} | \mathbf{D})$ denote the ATT conditional on $\mathbf{D}$, and let $\tilde{\beta}_{fe} = E(\hat{\beta}_{fe} | \mathbf{D})$. The first measure we consider is the minimal value of the standard deviation of the $\tilde{\Delta}_{g,t}$ s under which one could have that $\tilde{\beta}_{fe}$ is of a different sign than $\tilde{\Delta}^{TR}$. Therefore, this summary measure applies to $\tilde{\beta}_{fe}$ and $\tilde{\Delta}^{TR}$, rather than β_{fe} and δ^{TR} , the unconditional expectations of $\hat{\beta}_{fe}$ and Δ^{TR} on which we have focused so far. However, one can show that when G , the number of groups, goes to infinity, $\tilde{\beta}_{fe} - \beta_{fe}$ and $\tilde{\Delta}^{TR} - \delta^{TR}$ both converge to 0. So if the number of groups is large, $\tilde{\beta}_{fe}$ and $\tilde{\Delta}^{TR}$ should not differ much from β_{fe} and δ^{TR} , and our robustness measure “almost” applies to β_{fe} and δ^{TR} .

Let

$$\sigma(\tilde{\Delta}) = \left(\sum_{(g,t): D_{g,t}=1} \frac{N_{g,t}}{N_1} (\tilde{\Delta}_{g,t} - \tilde{\Delta}^{TR})^2 \right)^{1/2},$$

$$\sigma(\mathbf{w}) = \left(\sum_{(g,t): D_{g,t}=1} \frac{N_{g,t}}{N_1} (w_{g,t} - 1)^2 \right)^{1/2}.$$

¹⁰Borusyak and Jaravel (2017) assume that the treatment effect of cell (g, t) only depends on the number of periods since group g has started receiving the treatment, whereas Proposition 1 does not rely on that assumption.

¹¹ $\tilde{\Delta}_{g,t}$ may differ from $E(\Delta_{g,t})$. To see this, let us consider a simple example where $T = 2$. Then, under Assumption 3, one has $\tilde{\Delta}_{g,t} = E(\Delta_{g,t} | D_{g,1}, D_{g,2})$. One may for instance have $E(\Delta_{g,1} | D_{g,1} = 0, D_{g,2} = 0) < E(\Delta_{g,1} | D_{g,1} = 1, D_{g,2} = 1)$, if a group is more likely to be treated if her treatment effect is initially high.

$\sigma(\tilde{\Delta})$ is the standard deviation of the conditional ATEs, and $\sigma(\mathbf{w})$ is the standard deviation of the $\mathbf{w}$ -weights,¹² across the treated (g, t) cells. Let $n = \#\{(g, t) : D_{g,t} = 1\}$ denote the number of treated cells. For every $i \in \{1, \dots, n\}$, let $w_{(i)}$ denote the i th largest of the weights of the treated cells: $w_{(1)} \geq w_{(2)} \geq \dots \geq w_{(n)}$, and let $N_{(i)}$ and $\tilde{\Delta}_{(i)}$ be the number of observations and the conditional ATE of the corresponding cell. Then, for any $k \in \{1, \dots, n\}$, let $P_k = \sum_{i \geq k} N_{(i)} / N_1$, $S_k = \sum_{i \geq k} (N_{(i)} / N_1) w_{(i)}$ and $T_k = \sum_{i \geq k} (N_{(i)} / N_1) w_{(i)}^2$.

Corollary 1 *Suppose that Assumptions 1-5 hold.*

1. *If $\sigma(\mathbf{w}) > 0$, the minimal value of $\sigma(\tilde{\Delta})$ compatible with $\tilde{\beta}_{fe}$ and $\tilde{\Delta}^{TR} = 0$ is*

$$\underline{\sigma}_{fe} = \frac{|\tilde{\beta}_{fe}|}{\sigma(\mathbf{w})}.$$

2. *If $\tilde{\beta}_{fe} \neq 0$ and at least one of the $w_{g,t}$ weights is strictly negative, the minimal value of $\sigma(\tilde{\Delta})$ compatible with $\tilde{\beta}_{fe}$ and with $\tilde{\Delta}_{g,t}$ of a different sign than $\tilde{\beta}_{fe}$ for all (g, t) is*

$$\underline{\sigma}_{fe} = \frac{|\tilde{\beta}_{fe}|}{[T_s + S_s^2 / (1 - P_s)]^{1/2}},$$

where $s = \min\{i \in \{1, \dots, n\} : w_{(i)} < -S_{(i)} / (1 - P_{(i)})\}$.

$\underline{\sigma}_{fe}$ and $\underline{\sigma}_{fe}$ can be estimated simply by replacing $\tilde{\beta}_{fe}$ by $\hat{\beta}_{fe}$. An estimator of $\underline{\sigma}_{fe}$ can be used to assess the robustness of $\hat{\beta}_{fe}$ to treatment effect heterogeneity across groups and periods. If $\underline{\sigma}_{fe}$ is close to 0, $\tilde{\beta}_{fe}$ and $\tilde{\Delta}^{TR}$ can be of opposite signs even under a small and plausible amount of treatment effect heterogeneity. In that case, treatment effect heterogeneity would be a serious concern for the validity of $\hat{\beta}_{fe}$. On the contrary, if $\underline{\sigma}_{fe}$ is very large, $\tilde{\beta}_{fe}$ and $\tilde{\Delta}^{TR}$ can only be of opposite signs under a very large and implausible amount of treatment effect heterogeneity. Then, treatment effect heterogeneity is less of a concern.

Similarly, if $\underline{\sigma}_{fe}$ is close to 0, one may have, say, $\tilde{\beta}_{fe} > 0$, while $\tilde{\Delta}_{g,t} \leq 0$ for all (g, t) , even if the dispersion of the $\tilde{\Delta}_{g,t}$ s across (g, t) cells is relatively small. Notice that $\underline{\sigma}_{fe}$ is only defined if at least one of the weights is strictly negative: if all the weights are positive, then one cannot have that $\tilde{\beta}_{fe}$ is of a different sign than all the $\tilde{\Delta}_{g,t}$ s.

When some of the weights $w_{g,t}$ are negative, $\hat{\beta}_{fe}$ may still be robust to heterogeneous treatment effects across groups and periods, provided the assumption below is satisfied.

Assumption 7 ($\mathbf{w}$ uncorrelated with $\tilde{\Delta}$) $E \left[\sum_{(g,t): D_{g,t}=1} \frac{N_{g,t}}{N_1} (w_{g,t} - 1) (\tilde{\Delta}_{g,t} - \tilde{\Delta}^{TR}) \right] = 0$.

Corollary 2 *If Assumptions 1-5 and 7 hold, then $\beta_{fe} = \delta^{TR}$.*

¹² One can show that $\sum_{(g,t): D_{g,t}=1} (N_{g,t} / N_1) w_{g,t} = 1$.

Assumption 7 requires that the weights attached to the fixed effects estimator be uncorrelated with the conditional ATEs in the treated (g, t) cells. This is often implausible. For instance, groups treated the most are also those with the lowest value of $w_{g,t}$, as shown in Proposition 1. But those groups could also be those with the largest treatment effect. This would then induce a negative correlation between $\mathbf{w}$ and $\tilde{\Delta}$. The plausibility of Assumption 7 can be assessed, by looking at whether $\mathbf{w}$ is correlated with a predictor of the treatment effect in each (g, t) cell. In the two applications we revisit in Section 6, this test is rejected.

3.3 Extension to the first-difference regression

Instead of Regression 1, many articles have estimated the first-difference regression defined below:

Regression 2 (*First-difference regression*)

Let $\hat{\beta}_{fd}$ denote the coefficient of $D_{g,t} - D_{g,t-1}$ in an OLS regression of $Y_{g,t} - Y_{g,t-1}$ on period fixed effects and $D_{g,t} - D_{g,t-1}$, among observations for which $t \geq 2$. Let $\beta_{fd} = E \left[\hat{\beta}_{fd} \right]$.

When $T = 2$ and $N_{g,2}/N_{g,1}$ does not vary across g , meaning that all groups experience the same growth of their number of units from period 1 to 2, one can show that $\hat{\beta}_{fe} = \hat{\beta}_{fd}$. $\hat{\beta}_{fe}$ differs from $\hat{\beta}_{fd}$ if $T > 2$ or $N_{g,2}/N_{g,1}$ varies across g .

We start by showing that a result similar to Theorem 1 also applies to $\hat{\beta}_{fd}$. For any $(g, t) \in \{1, \dots, G\} \times \{2, \dots, T\}$, let $\varepsilon_{fd,g,t}$ denote the residual of observations in group g and at period t in the regression of $D_{g,t} - D_{g,t-1}$ on period fixed effects, among observations for which $t \geq 2$. For any $g \in \{1, \dots, G\}$, let $\varepsilon_{fd,g,1} = \varepsilon_{fd,g,T+1} = 0$. One can show that if the regressors in Regression 2 are not perfectly collinear,

$$\sum_{(g,t):D_{g,t}=1} \frac{N_{g,t}}{N_1} \left(\varepsilon_{fd,g,t} - \frac{N_{g,t+1}}{N_{g,t}} \varepsilon_{fd,g,t+1} \right) \neq 0.$$

Then we define

$$w_{fd,g,t} = \frac{\varepsilon_{fd,g,t} - \frac{N_{g,t+1}}{N_{g,t}} \varepsilon_{fd,g,t+1}}{\sum_{(g,t):D_{g,t}=1} \frac{N_{g,t}}{N_1} \left(\varepsilon_{fd,g,t} - \frac{N_{g,t+1}}{N_{g,t}} \varepsilon_{fd,g,t+1} \right)}.$$

Theorem 2 *Suppose that Assumptions 1-5 hold. Then,*

$$\beta_{fd} = E \left[\sum_{(g,t):D_{g,t}=1} \frac{N_{g,t}}{N_1} w_{fd,g,t} \Delta_{g,t} \right].$$

Theorem 2 shows that under Assumption 5, β_{fd} is equal to a weighted sum of the ATEs in each treated (g, t) cell with potentially some strictly negative weights, just as β_{fe} . We now characterize the (g, t) cells whose ATEs are weighted negatively by β_{fd} . To do so, we focus on